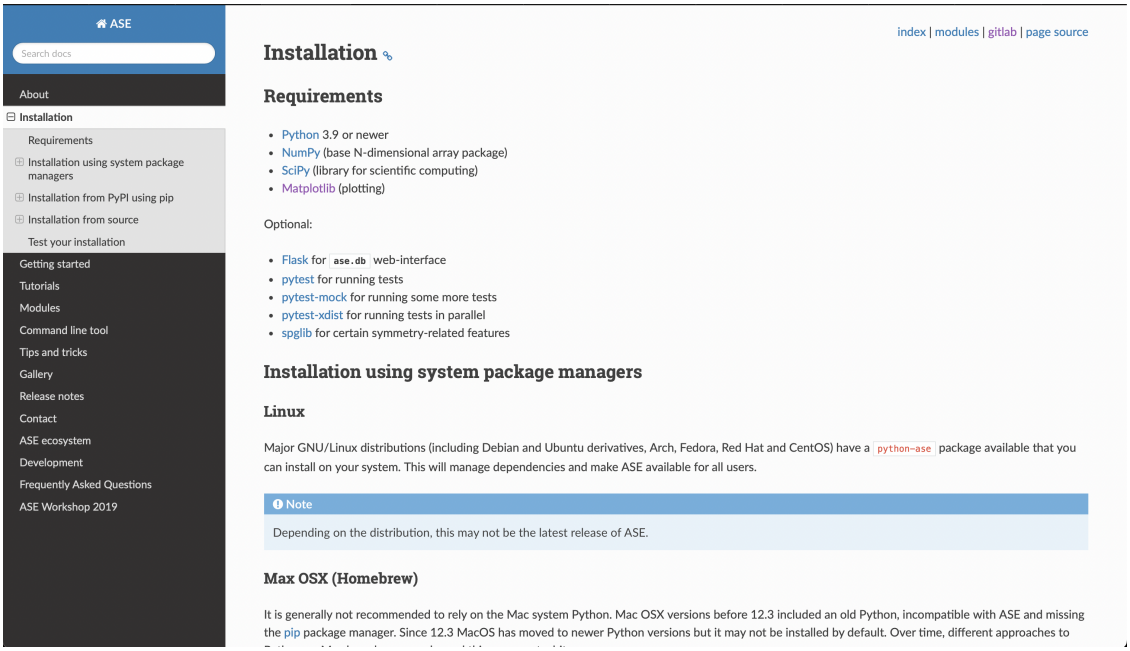
ASE Introduction

October 29, 2025

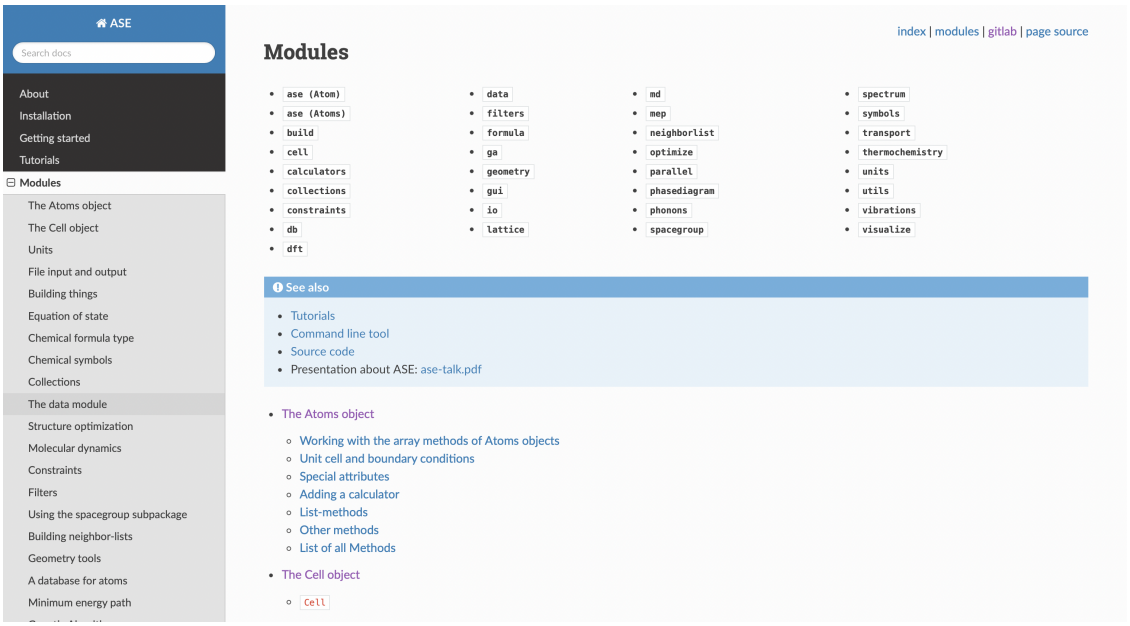
1 The Atomic Simulation Environment - An Introduction

1.1 What is ASE?

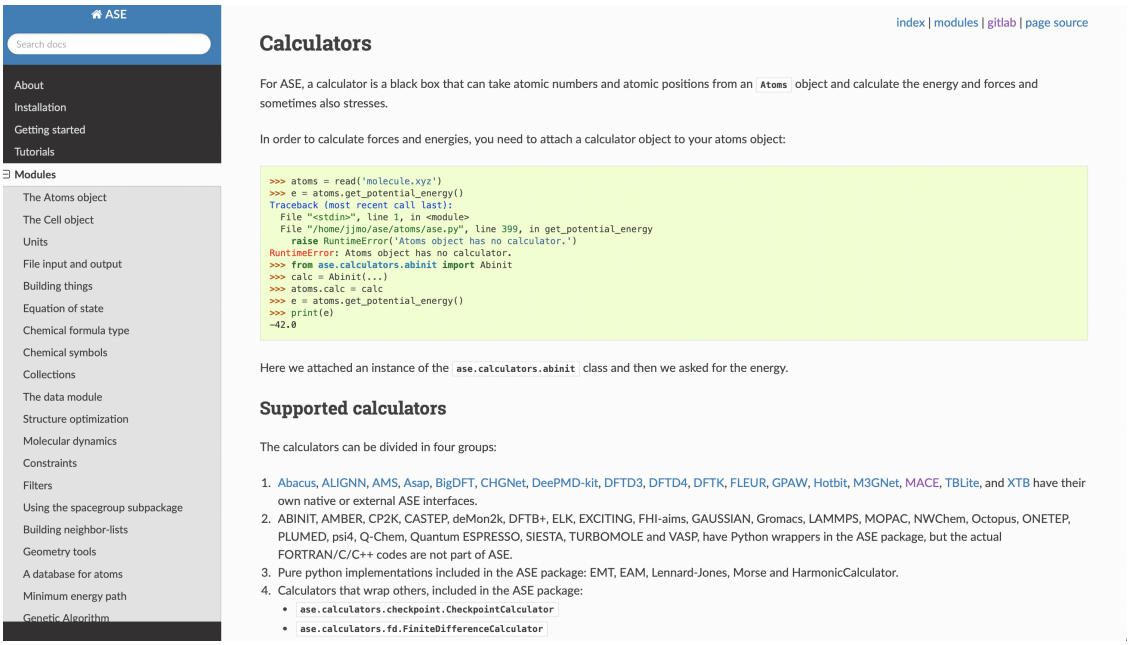
ASE is Python library for the atomistic simulation of molecules and materials. https://wiki.fysik.dtu.dk/ase/index.html



It contains modules for working with atoms

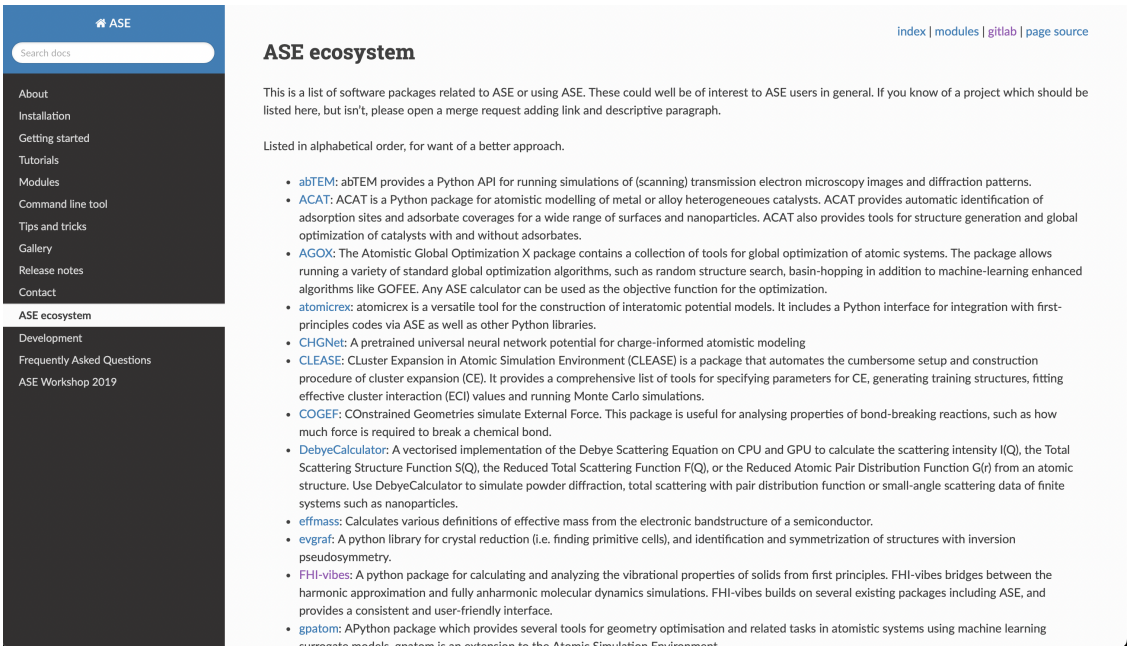


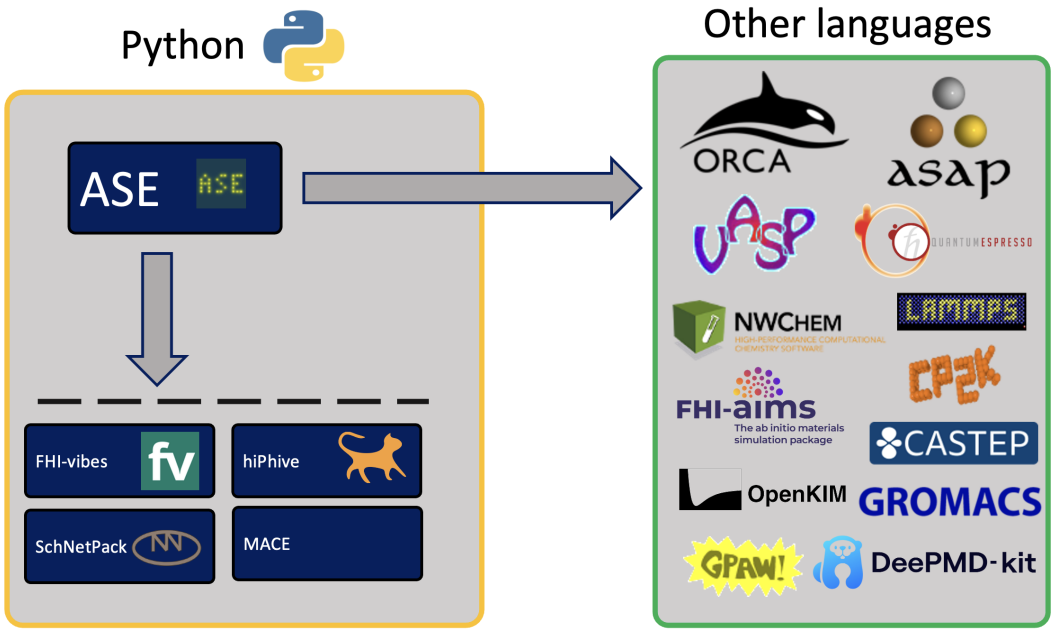
Most importantly is the calculator module



This means that ASE is very flexible and can interface with different codes and ultimatley different levels of theory

Since ASE is written in Python, it also interfaces with other Python codes for materials modelling, which creates an ecosystem of software and enhances our scientific toolbox.





We can build all sorts of structures with ASE from molecules to crystals

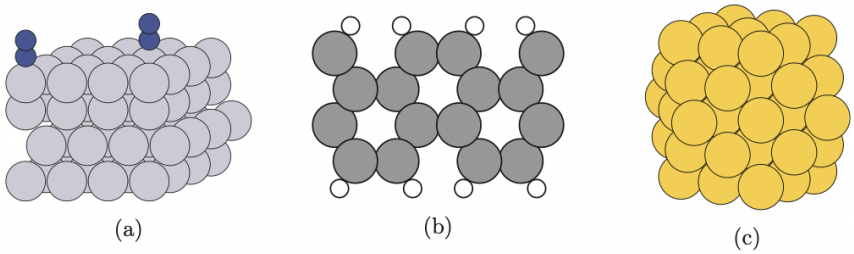
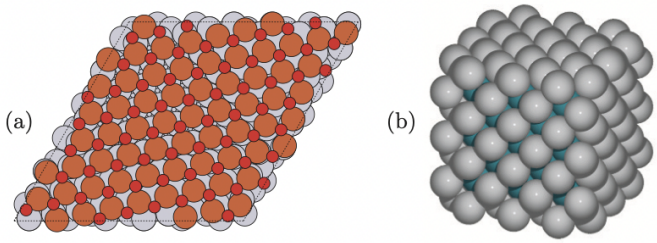


Figure 2. (a) Platinum surface with 2 N_2 adsorbed at top sites [55], (b) carbon nanoribbon with H-saturated edge, and (c) cuboctahedral gold nanoparticle constructed using various functions of ASE.



1.2 OK but how do we actually use ASE to model materials?

We can do a lot of things with ASE. We can use it to:

- Build materials
- Simulate materials (Vibrations, energy calculations, molecular dynamics, transition states)
- Visualise simulations

Let's take a look at an example calculation

[2]:
```
from ase import Atoms
from ase.calculators.emt import EMT
from ase.visualize import view

atom = Atoms('N')
atom.calc = EMT()
e_atom = atom.get_potential_energy()

d = 1.1
molecule = Atoms('2N', [(0., 0., 0.), (0., 0., d)])
molecule.calc = EMT()
e_molecule = molecule.get_potential_energy()

e_atomization = e_molecule - 2 * e_atom

print('Nitrogen atom energy: %5.2f eV' % e_atom)
print('Nitrogen molecule energy: %5.2f eV' % e_molecule)
print('Atomization energy: %5.2f eV' % -e_atomization)
```

```
view(molecule, viewer="ngl")
```

```
Nitrogen atom energy:  5.10 eV
Nitrogen molecule energy:  0.44 eV
Atomization energy:  9.76 eV
```

```
HBox(children=(NGLWidget(), VBox(children=(Dropdown(description='Show',␣
↪options=('All', 'N'), value='All'), Dr...
```

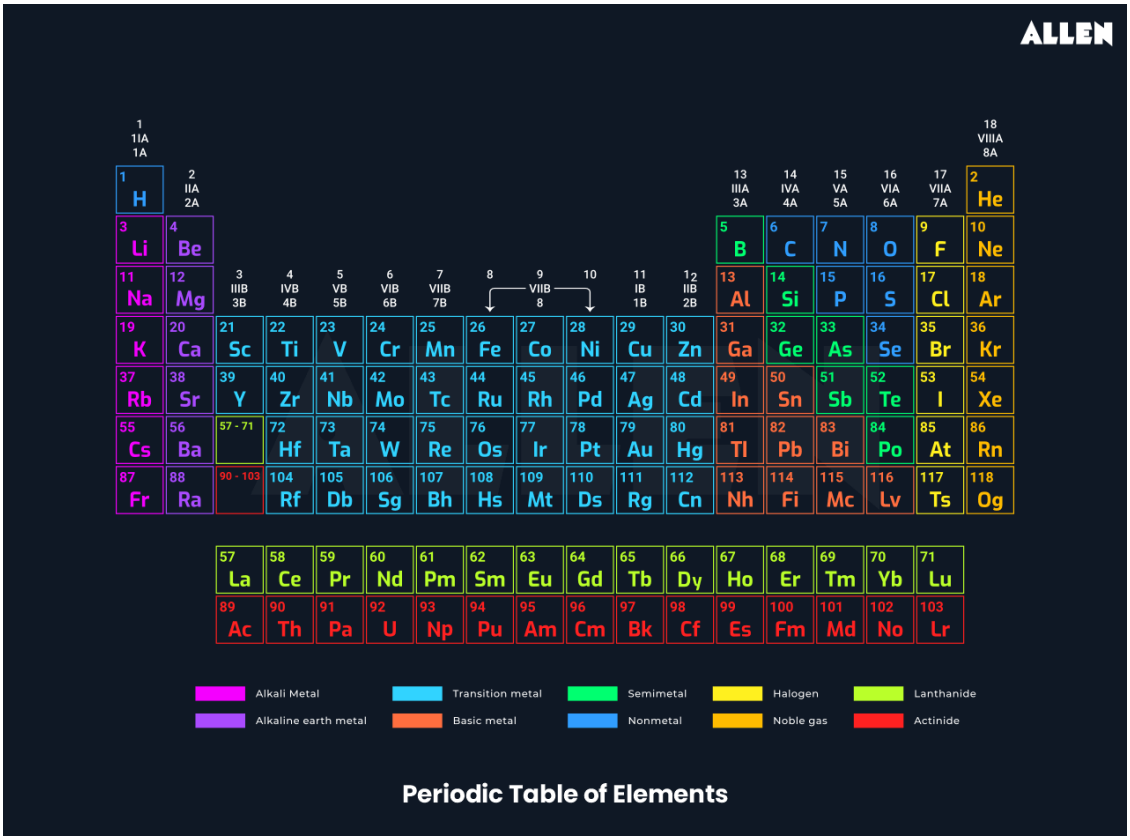
[4]:
```
from ase.build import nanotube
cnt1 = nanotube(6, 0, length=4)
cnt2 = nanotube(3, 3, length=6, bond=1.4, symbol='Si')
view(cnt1, viewer="ngl")
```

```
HBox(children=(NGLWidget(), VBox(children=(Dropdown(description='Show',␣
↪options=('All', 'Si'), value='All'), D...
```

[6]:
```
import numpy as np

from ase import Atoms
from ase.build import molecule
from ase.io import write

a = 5.64  # Lattice constant for NaCl
cell = [a / np.sqrt(2), a / np.sqrt(2), a]
atoms = Atoms(symbols='Na2Cl2', pbc=True, cell=cell,
              scaled_positions=[(.0, .0, .0),
                                (.5, .5, .5),
                                (.5, .5, .0),
                                (.0, .0, .5)]) * (3, 4, 2) + molecule('C6H6')

# Move molecule to 3.5Ang from surface, and translate one unit cell in xy
atoms.positions[-12:, 2] += atoms.positions[:-12, 2].max() + 3.5
atoms.positions[-12:, :2] += cell[:2]

# Mark a single unit cell
atoms.cell = cell

view(atoms, viewer="ngl")
```

```
HBox(children=(NGLWidget(), VBox(children=(Dropdown(description='Show',␣
↪options=('All', 'Cl', 'H', 'Na', 'C'),...
```

Once we've greated them we can start playing around with them. We have the method and now we can crate anything from any ingredients


ALLEN
1 1IA 1A
2 IIA 2A
3 IIIB 3B
4 IVB 4B
5 VB 5B
6 VIB 6B
7 VIIB 7B
8
9 VIIB 8
10
11 IB 1B
12 IIB 2B
13 IIIA 3A
14 IVA 4A
15 VA 5A
16 VIA 6A
17 VIIA 7A
18 VIIIA 8A
H He
Li Be B C N O F Ne
Na Mg Al Si P S Cl Ar
K Ca Sc Ti V Cr Mn Fe Co Ni Cu Zn Ga Ge As Se Br Kr
Rb Sr Y Zr Nb Mo Tc Ru Rh Pd Ag Cd In Sn Sb Te I Xe
Cs Ba 57 - 71 Hf Ta W Re Os Ir Pt Au Hg Tl Pb Bi Po At Rn
Fr Ra 90 - 103 Rf Db Sg Bh Hs Mt Ds Rg Cn Nh Fi Mc Lv Ts Og
La Ce Pr Nd Pm Sm Eu Gd Tb Dy Ho Er Tm Yb Lu
Ac Th Pa U Np Pu Am Cm Bk Cf Es Fm Md No Lr
Alkali Metal
Alkaline earth metal
Transition metal
Basic metal
Semimetal
Nonmetal
Halogen
Noble gas
Lanthanide
Actinide
Periodic Table of Elements

[ ]: